

REVIEW OF PREVIOUS LECTURE

- If X, Y **independent** random variables:

$$E(XY) = E(X)E(Y), \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y), \text{Cov}(X, Y) = 0$$

- Bernoulli random variable: $X \sim \text{Bernoulli}(p)$

$$P(X = 1) = p \quad P(X = 0) = 1 - p$$

$$E(X) = p, \text{Var}(X) = p(1 - p)$$

- Binomial random variable: $X \sim \text{Binomial}(n, p)$

$$\binom{n}{x} p^x (1 - p)^{n-x} \quad E(X) = np, \text{Var}(X) = np(1 - p)$$

WIDELY USED DISCRETE DISTRIBUTIONS: HYPERGEOMETRIC

X has Hypergeometric distribution with parameters n, A, B , if

$$f(x) = \begin{cases} \frac{\binom{A}{x} \binom{B}{n-x}}{\binom{A+B}{n}} & \text{for } \max(0, n-B) \leq x \leq \min(n, A) \\ 0 & \text{otherwise} \end{cases}$$

- **Example:** A red, B blue balls.
 n balls selected randomly without replacement.
 X : # of red balls selected.

WIDELY USED DISCRETE DISTRIBUTIONS: HYPERGEOMETRIC

- **Expectation** of Hypergeometric distribution:

Let $p = \frac{A}{A+B}$. Let $X_i = 1$ if i^{th} ball red, $X_i = 0$ otherwise.

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Although X_i are dependent, we have

$$P(X_i = 1) = p, P(X_i = 0) = 1 - p \qquad E(X_i) = p$$

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$$E(X) = \sum_{i=1}^n E(X_i) = \sum_{i=1}^n p = np$$

Same as Binomial Distribution (Sampling with replacement).

WIDELY USED DISCRETE DISTRIBUTIONS: HYPERGEOMETRIC

- **Variance** of Hypergeometric distribution:

We will not derive it. Since X_i dependent, as expected, variance will differ from that of Binomial:

$$Var(X) = np(1 - p) \left(\frac{T - n}{T - 1} \right)$$

$T = A + B$ is size of population, $p = \frac{A}{A + B}$, n is sample size

Variance smaller than that of Binomial (Sampling with replacement).

WIDELY USED DISCRETE DISTRIBUTIONS: HYPERGEOMETRIC

- Comparison of Binomial and Hypergeometric:

Although their pmf and variances are different:

If Population size $A + B \gg$ Sample size $n \Rightarrow$

Sampling with replacement $\equiv$ Sampling without replacement

(Binomial distr. almost same as Hypergeometric distr.)

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(Binomial distr. almost same as Hypergeometric distr.)

- **How big?** n is less than 5 % of population size.

```
from scipy.stats import binom  
binom.pmf(0, 100, 0.002)
```

0.8185668046884276

```
from scipy.stats import hypergeom  
hypergeom.pmf(0, 2000, 4, 100)
```

0.8143774997343618

WIDELY USED DISCRETE DISTRIBUTIONS: POISSON

X has Poisson distribution with parameter λ , if

$$f(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!} & \text{for } x = 0, 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

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Can model occurrences of random arrivals:

Customer arrivals, traffic accidents, flood occurrences, etc.

Assume # of arrivals in any interval is independent.

Used to approximate Binomial if large n , small p (rare event).

Note: Both expectation and variance = λ .

WIDELY USED DISCRETE DISTRIBUTIONS: POISSON

Ex: Rare disease with probability 0.01.

Probability that in a random group of 200, 3 people have disease

Let X : # of people with disease.

With Binomial (exact): $P(X = 3) = \binom{200}{3} (0.01)^3 (0.99)^{197} \approx 0.181$

With Poisson (approximation):

$$\lambda = 200 \times 0.01 = 2 \quad P(X = 3) = e^{-2} \frac{2^3}{3!} \approx 0.18$$

WIDELY USED DISCRETE DISTRIBUTIONS: POISSON

Ex: Customers arrive at a store at a rate of 3/hour on average.
Assume customer arrivals in different time periods independent.
Probability that 10 or more customers arrive between 2pm-4pm

of customers in any 2 hour period, $\lambda = 6$.

```
from scipy.stats import poisson  
print(1-poisson.cdf(9, 6))
```

```
0.08392401699487584
```

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

X has Normal (Gaussian) distribution with parameter μ, σ^2 , if

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Denoted with $X \sim N(\mu, \sigma^2)$

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

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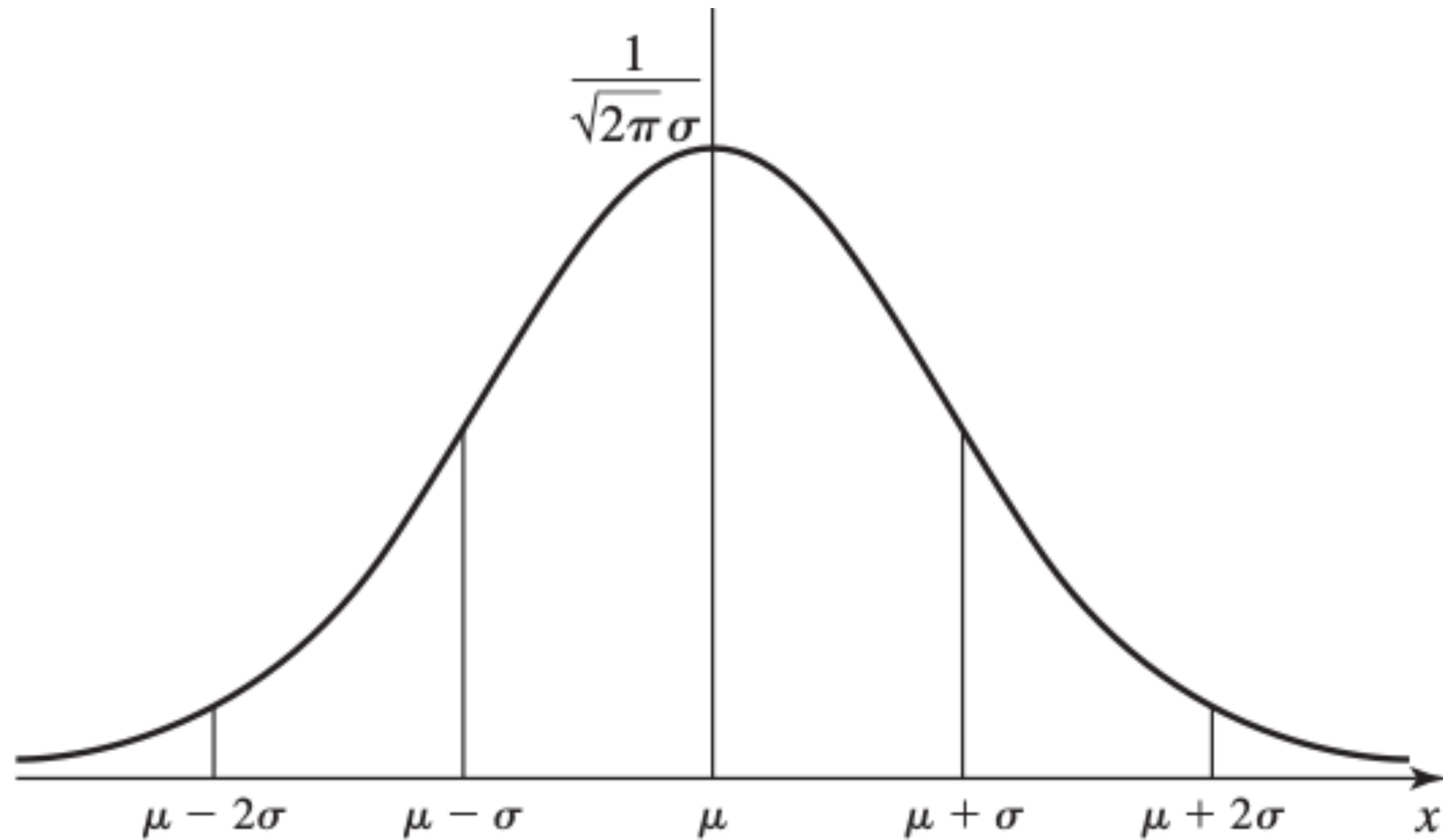
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Denoted with $X \sim N(\mu, \sigma^2)$

Most common distribution:

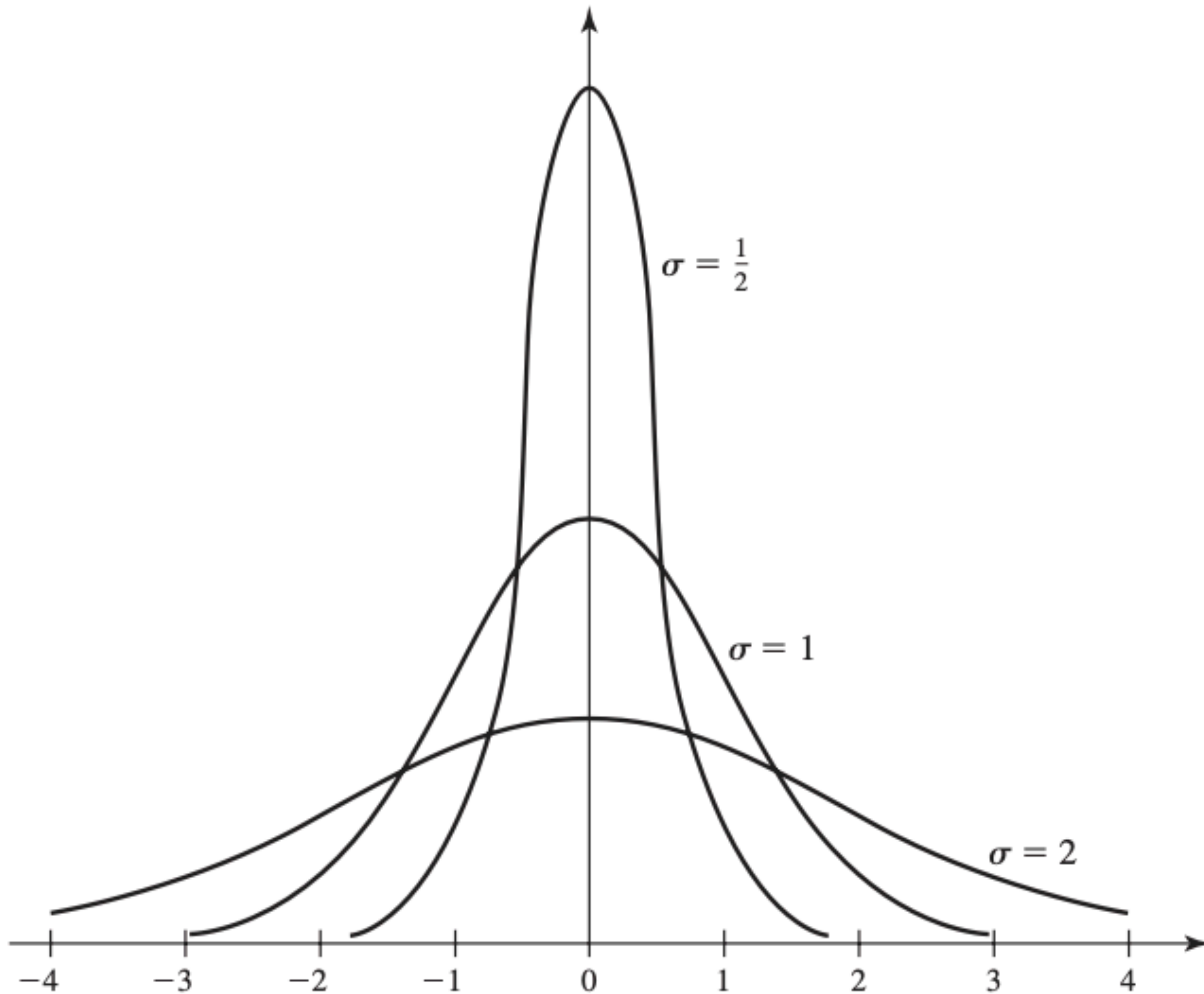
- Mathematical convenience
- Many physical phenomena approximately normal
- Central Limit Theorem: Mean of Sample from **any distribution** is approximately normal (More in a little while)

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL



Symmetric around mean

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL



As σ grows,
pdf gets more flat

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

Linear Transformations of normal independent variables:

$$X_1 \sim N(\mu_1, \sigma_1^2) \text{ and}$$

$$X_2 \sim N(\mu_2, \sigma_2^2)$$

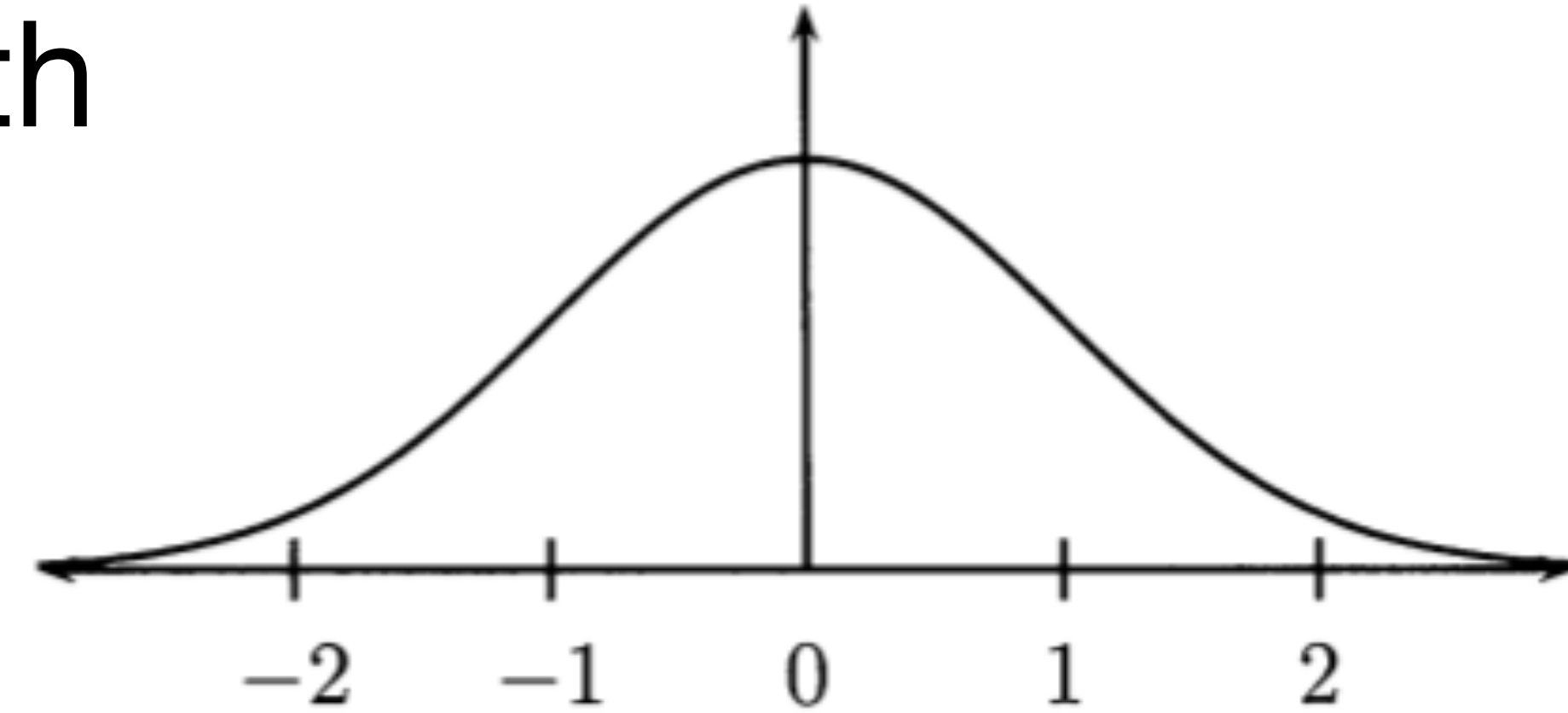
$$\Rightarrow Y = a_1X_1 + a_2X_2 + b \sim N(a_1\mu_1 + a_2\mu_2 + b, a_1^2\sigma_1^2 + a_2^2\sigma_2^2)$$

(Can be generalized to $k > 2$ random variables)

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

Standard Normal: Normal distribution with

$$\mu = 0, \sigma^2 = 1$$



- Denoted with $Z \sim N(0,1)$.
- Pdf denoted with $\phi(z)$. Cdf denoted with $\Phi(z)$.

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

From normal to standard normal:

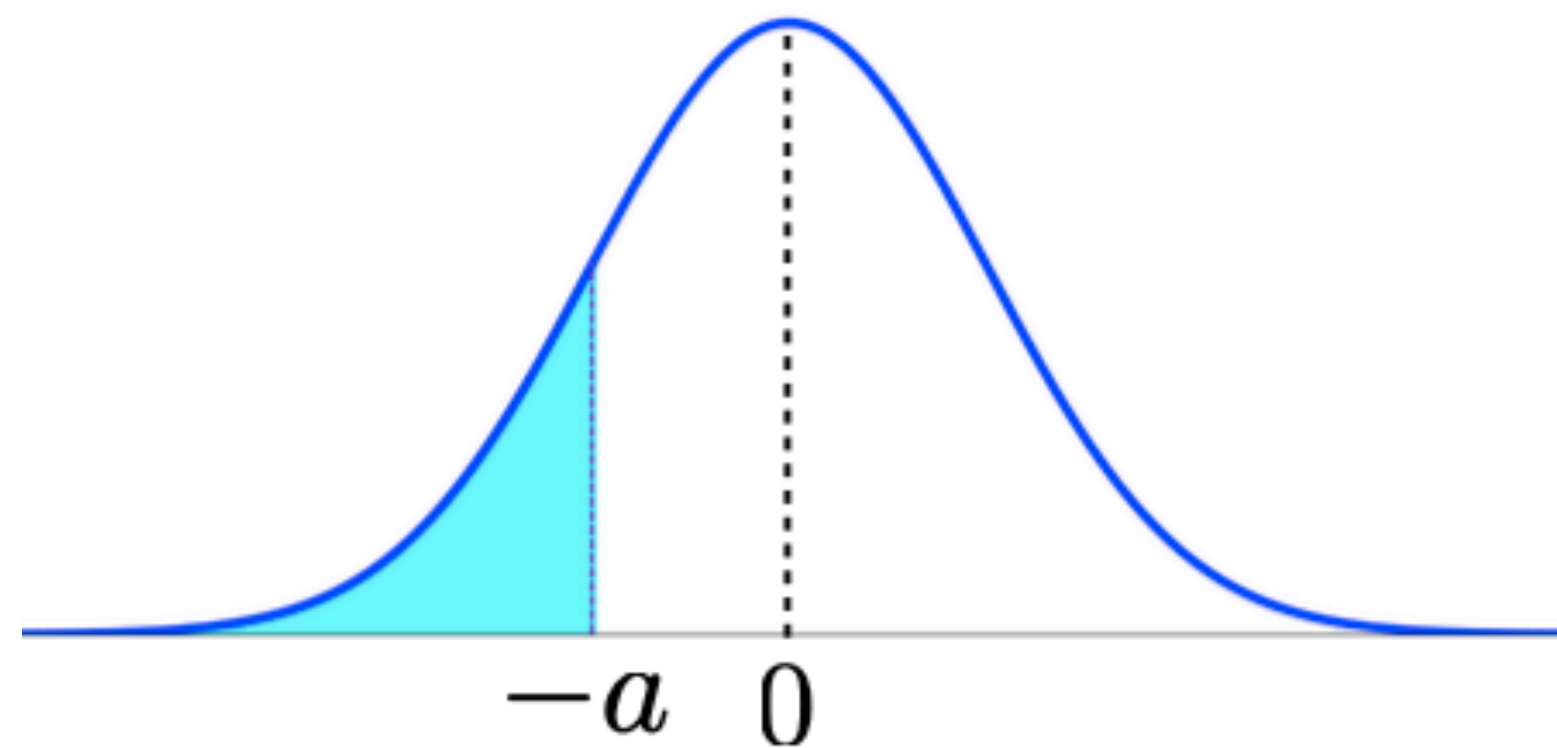
$$X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0,1)$$

(Follows from linear transformations of normal independent RVs)

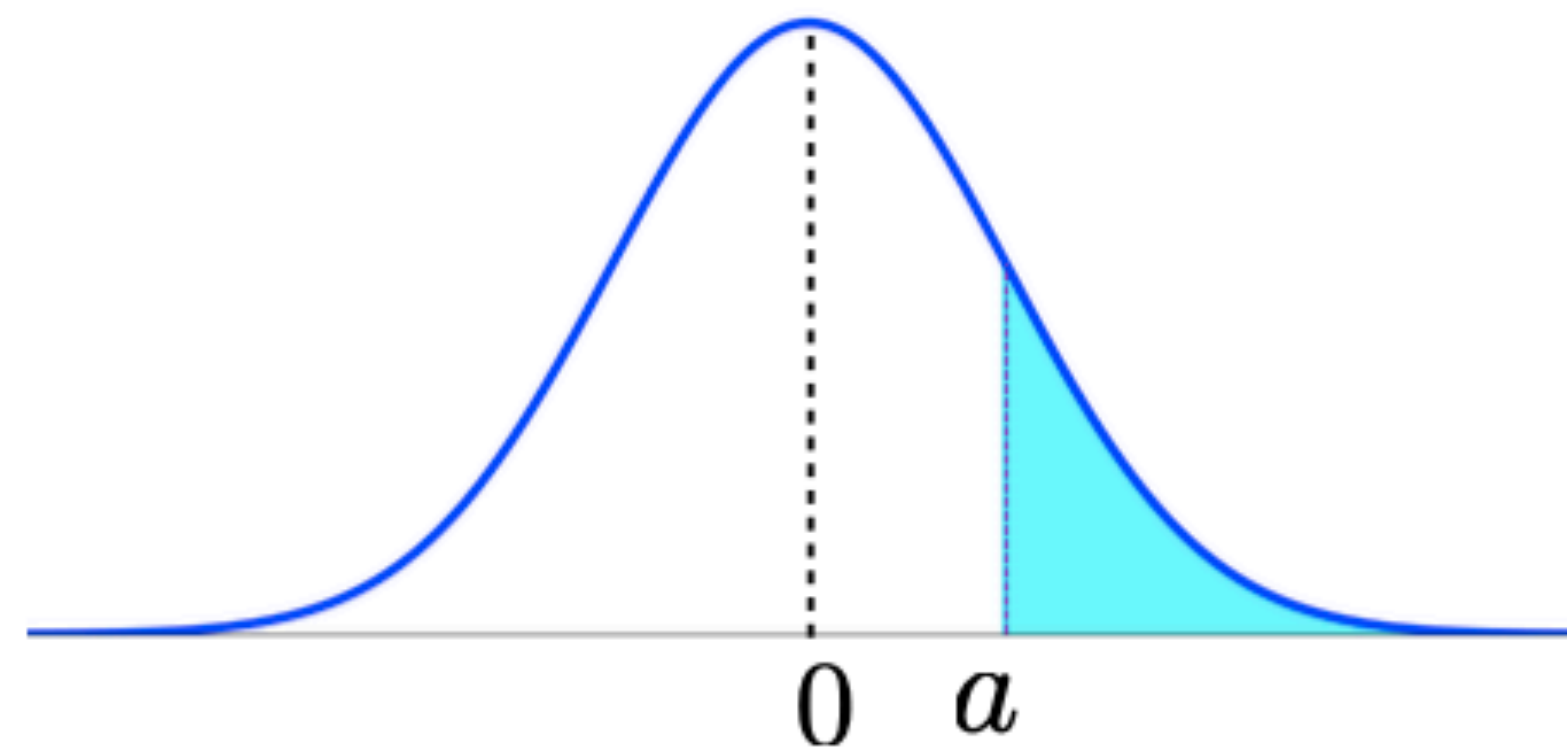
$$\begin{aligned} \Rightarrow P(a < X < b) &= P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right) \\ &= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right) \end{aligned}$$

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

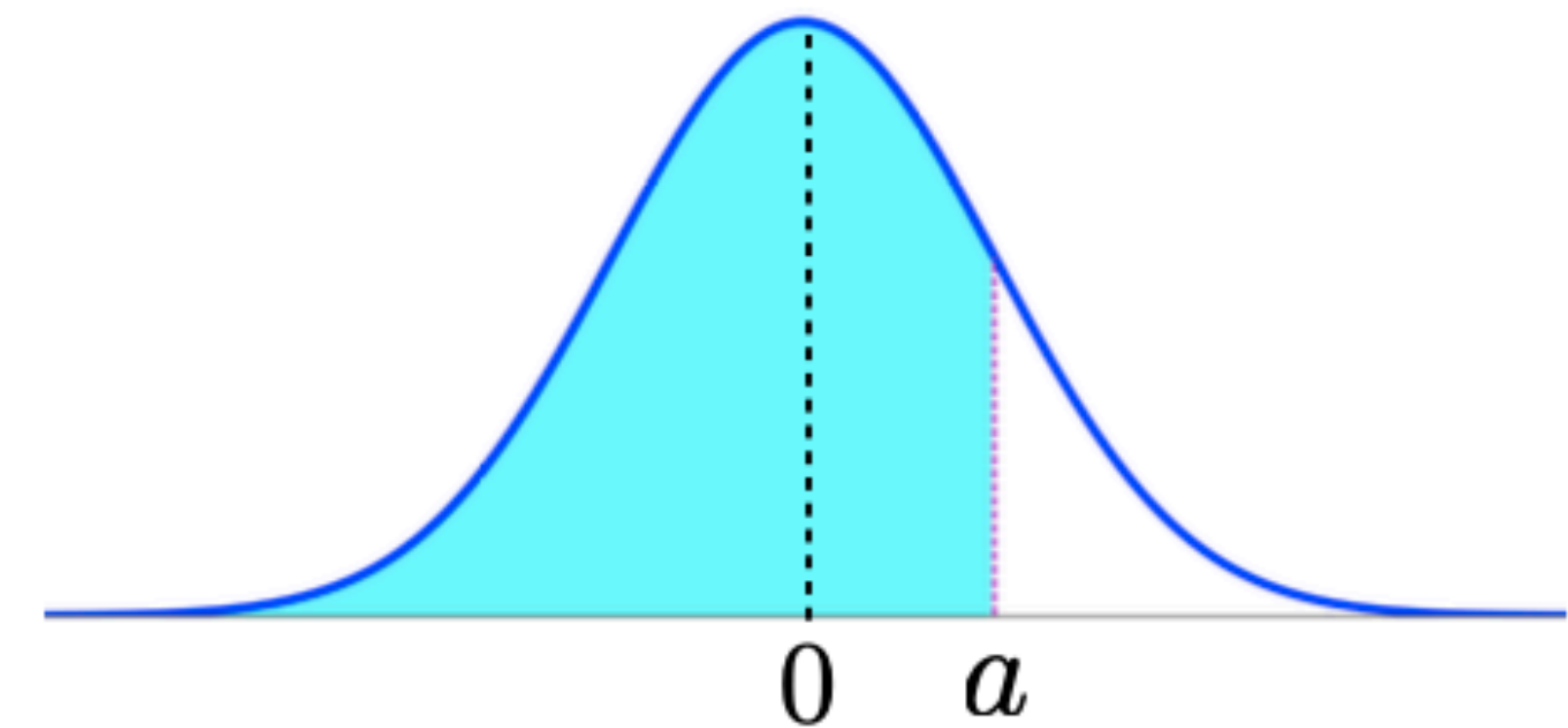
Standard Normal: Consequence of symmetry



$$\Phi(-a) = P(Z \leq -a)$$



$$= P(Z \geq a)$$



$$= 1 - P(Z \leq a) = 1 - \Phi(a)$$

WIDELY USED CONTINUOUS DISTRIBUTIONS: NORMAL

Ex: $X \sim N(5,4) \Rightarrow P(1 < X < 8) = ?$

$$\begin{aligned} P(1 < X < 8) &= P\left(\frac{1-5}{2} < \frac{X-5}{2} = Z < \frac{8-5}{2}\right) \\ &= \Phi(1.5) - \Phi(-2) = \Phi(1.5) - (1 - \Phi(2)) \end{aligned}$$

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```
from scipy.stats import norm  
print(norm.cdf(1.5)-(1-norm.cdf(2)))
```

0.9104426667829627